



FUEL MANAGEMENT Department
SIMHADRI

Note No :1



Ref:SMTTP/Fuel Mgmt./2023-24/LMI/CHP/636494

Date:
08/11/2023(10:43:23)

Subject:Approval for LMI on Performance and Capacity test Equipment” w.r.to revised OGN.

CC-OS has issued a revised OGN on “Performance and Capacity Test of Coal handling plant” (COS-ISO-00-OGN/OPS/PERF/009) Rev. 3 (attached as annexure 1).

Based on the above, CHP has reviewed existing LMI Ref. No. “LMI/OGN/OPS/PERF/009 Rev.02 (attached as annexure 2) and it is observed that LMI is in line with revised CC-OS OGN except the following’s exceptions:

1. RBFs (RBF 1, RBF 2, RBF 3) path capacity limited shall be limited to 75% of the rated capacity (it is already approved in the existing LMI).
2. Conducting of Capacity Test is exempted in the monsoon period (Between June 1st to Oct 31st) (it is already approved in the existing LMI).
3. “Tripper overspeed protection” is not available. In all MTs, drag protection has been installed.

Incorporating the above exceptions, Proposed LMI is hereby attached as annexure 3.

Being station LMI, the Competent Authority for approval of LMI shall be Head of Project.

Submitted for kind approval please.

Submitted by:

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Note No:2

Proposed LMI (Annexure 3) may be approved

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Note No:3

LMI reviewed , May be approved.

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Note No:7

Please attach only two docs in the annexure, one the **latest OGN** and two the **LMI in revised form in .pdf** that is to be accorded approval. Too many docs in the Annexures will only confuse the approver. Please review this case file in this light.

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Note No:8

May pl review

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Note No:9

Annexure 1- Revised OGN issued by COS in Sep'23

Annexure 2- Existing LMI

Annexure 3 -Revised LMI being proposed

File nomenclature is changed for clarity.

Submitted pl

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Note No:11

Please change the details given in the 'footer' of proposed LMI from *COS-ISO-00-OGN/OPS/PERF/009* to **LMI/OGN/OPS/PERF/009** to represent the LMI document itself.

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Note No:12

Necessary correction in proposed LMI is done. Submitted pl

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Note No:13

- 1) Please include the approval page also (see page no. 2 of 16 of existing LMI i.e. Rev. no. 2 - copy attached in annexures). Kindly see to it that the names and designations in the revised approval page, are correct.
- 2) On page no. 15 of 16 of existing LMI, clause no.7, Review of LMI was done by BUH, and on page no. 2 of 9 review approval for the OGN is being accorded by head of COS i.e. ED(OS), hence please change the review approval of the revised LMI, (in clause no. 1.3) as BUH (Simhadri).

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Note No:15

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Note No:16

Necessary changes as advised have been made.

May please approve.

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Note No:18

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Note No:19

May be approved please.

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Note No:20

May please approve.

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On:27/11/2023(18:40:52) **Ref:**SMTTP/Fuel Mgmt./2023-24/LMI/CHP/636494



LMI

SIMHADRI

TITLE: PERFORMANCE & CAPACITY TEST OF COAL HANDLING PLANT

REF NO: LMI/OGN/OPS/PERF/009

REVISION: 03

(For Internal circulation only)

REF DOCUMENT: COS-ISO-00-OGN/OPS/PERF/009) Rev. 3, Date: September 2023

LMI/OGN/OPS/PERF/009

LMI

Performance & Capacity Test of
Coal Handling Plant

Approved for implementation by:

.....

GGM, Simhadri

Date:

LMI APPROVAL DOCUMENT

1	TITLE OF LMI	PERFORMANCE & CAPACITY TEST OF COAL HANDLING PLANT
2	DOCUMENT NUMBER REVISION/ ISSUE NO & DATE	<u>LMI/OGN/OPS/PERF/009 REV. – 03</u> <u>Nov.'2023</u>
3	ORIGINAL/ REF DOCUMENT	COS-ISO-00-OGN/OPS/PERF/009) Rev. 3 Sep 2023
4	PREPARED BY	SIGNATURE: L P Shakya, DGM (CHP-OPN)
	CHECKED BY	SIGNATURE: Niraj Kumar, AGM (CHP-OPN)
5	CHECKED BY	SIGNATURE: A.R. Koteswara Rao, AGM (CHP-O&M)
6	RECHECKED BY	SIGNATURE: Sanjit Mazumdar, GM (FM)
7	REVIEWED BY	SIGNATURE: D P Patra, GM (O&M)
8	APPROVED FOR ISSUE	SIGNATURE: S.K Sinha, HOP(SSTPP)
9	DATE OF ISSUE	Nov'23

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1.0 SUMMARY

1.1 Superseded Document	LMI/OGN/OPS/PERF/009 Rev.02 Jul. 23	
1.2 Scope	Guidelines for Performance & Capacity Testing of Coal Handling Plant.	
1.3 Review	Review Responsibility	HoD (FM)
	Review Frequency	2 yearly basis or as and when required, whichever is lower
	Review Approval	BUH (Simhadri) shall be reviewing authority for any revision in this document

2.0 INTRODUCTION

The performance of the CHP (Conveyors & Major equipment) is to be monitored to improve efficiency and increase the availability and reliability of the coal Handling Plant. For ensuring best performance from this plant, the capabilities of conveyor system along with major equipment are to be measured. This document aims to describe the requirements, frequency, procedure, record keeping & reporting with respect to performance and capacity testing of coal handling plant.

3.0 RECORDS OF REVISION

1. Old Clause 1.1 & clause 6.0: Capacity Test Schedule and witnessing by Regional OS & Corporate OS has been modified from Four Per year (Quarterly) to 03 per year (01 in each quarter except July-September Quarter Owing to Monsoon) and witness by ROS once a year and by COS during yearly technical audit. For better clarity this point is specifically mentioned in clause 7.0 & 8.0 of the revised document.
2. Old Clause 1.1 & 1.3: The criteria of “power measurement” has been revised to “measurement of drive currents” and specified in clause 4 (ix) and clause 6.3(i) of the revised document.
3. Old Clause 1.1(vi): Flow Path Testing of Emergency Reclaim Hoppers have been removed owing to obsolescence & non provision in new designs.
4. Old Clause 4.2.6: A new point is added for incorporating “tripper over speed protection” in New Clause Ref 6 (xii) (g) of the revised document. “Tripper overspeed protection” is not available. In all MTs, drag protection has been installed.
5. Old Clause 4.2.11: “Performance Testing of Coal Sampling Unit” has been removed because coal sampling units are now not in use due to tripartite sampling at unloading end.
6. Old Clause 8.0: Exception Reporting has been modified as “Exceptions and actions taken are to be monitored in monthly ORT through CHP ORT Sheet & also in CC OS & RHQ-OS monthly structured meetings” in clause Ref 9.0 of the revised document.
7. Annexure-I: Sample CHP capacity test report has been added in Annexure-I as an illustration.

4.0 Pre-Requisite for Conducting Capability Test

- i. Belt Weigher: For measurement of flow capacity of a stream and individual equipment, weighing scales/belt weighers installed in the system shall be used. Belt weighers are to be calibrated before the test and recorded.
- ii. Sufficiently empty bunkers to facilitate continuous capacity feeding of coal during the test.
- iii. Sufficient space in stockyard during stacking & Sufficient heap of Coal during reclaiming of coal
- iv. Belt protection switches, local stop push buttons, hooters, brakes/rail clamps in working order.
- v. Adequate and uninterrupted water supply for equipment cooling & dust suppression.
- vi. Availability of plant communication system & fire detection & protection system.
- vii. Proper lubrication and oil level of all equipment.
- viii. Housekeeping of plant and removal of coal spillage at the start during Tests.
- ix. Current Measurements should be available in the control room.
- x. Proper Illumination.

5.0 Conveyor Flow Paths & Equipment for Demonstration of Performance Requirements

Capacity on Ton/Hr (*equivalent to 75 % of rated in RBF involved paths) of conveyor system including the intermediate equipment for each of the two parallel conveyor streams separately or any combination thereof is to be tested in the following flow path operations:

*Due to RBF constraint in the path, it is limited to 75% of the rated capacity (In the parts involving RBF-1, RBF-2 & RBF#3)

5.1 Flow Path I (Direct Stream – TH):

Paddle feeder taking feed of incoming coal from track hopper and discharging crushed coal into farthest bunker through travelling tripper on bunker conveyor including all intermediate conveyors and equipment.

5.2 Flow Path II (Direct Stream – WT):

Apron feeder taking feed of incoming coal from wagon tippler and discharging crushed coal into farthest bunker through travelling tripper on bunker conveyor including all intermediate Conveyors and equipment.

5.3 Flow Path III (Stacking Stream – TH):

Paddle feeder taking feed of incoming coal from track hopper and stacking coal in stockyard using yard conveyor with stacker/ reclaimer at farthest position and boom conveyor at highest angle including all intermediate equipment.

Flow Path IV (Stacking Stream – WT):

Apron feeder / Vibrating Feeder taking feed of incoming coal from wagon tippler and stacking coal in stockyard using yard conveyor with stacker/reclaimer at farthest position and boom conveyor at highest angle including all intermediate equipment.

5.5 Flow Path V (Reclaiming Stream):

Reclaiming coal from stockyard of CHP using yard conveyor with Stacker Reclaimer at farthest position from head end of yard conveyor with the boom conveyor at its lowest position and feeding last bunker of farthest unit through travelling trippers on bunker conveyor including all intermediate equipment.

The capacity & performance test of all major equipment which are a part of above flow paths (as applicable) like Paddle Feeders, Wagon Tippler, Side Arm Charger, Apron Feeder, Vibrating Screen/Grizzly Feeders, vibrating feeders, Crushers, Belt Feeders, Reversible belt feeders, stacker cum Reclaimers, Stackers, Reclaimers & travelling trippers shall also be done.

The capacity & performance test to be conducted for all the stages of CHP including interconnecting conveyors where multiple stages are there.

6.0 PROCEDURE FOR CAPACITY TEST

The tests shall be conducted as per the procedure specified as follows:

6.1 Flow Path Selection

- i. For each of the Flow Paths, two conveyor streams shall be formed with suitable CRISS-CROSS Combinations of 'A' or 'B' Conveyors/ intermediate equipment's.
- ii. All above streams formed shall be independently tested for performance & capacity. Further, the equipments/ conveyors (except common equipments) used for performance testing of one stream shall not be used for performance testing of the other stream.
- iii. In case of direct flow path, capacity & performance test should include operation of all the trippers.
- iv. While operating in Stacking path, Stacking will be done near the drive end of the yard conveyor so that the conveyor is tested under fully loaded conditions. While operating in reclaiming path, stacker / Reclaimer shall be near tail end of yard conveyor.
- v. While operating the capacity & performance of the flow paths, near central running of conveyors, non-spillage of coal and non-choking of chutes/ equipments shall be demonstrated which will be an indication of smooth continuous flow through the conveyor system.

6.2 Test Duration

Normal duration of the above capacity tests shall be at least one hour. A minimum of 30 minutes continuous & steady operation shall be required to establish the capacity, after flow rate stabilizes, then current readings will be measured.

In case of any interruption due to any reason, the duration of test shall be extended by the time lost in such interruptions.

6.3 Recording of Data

- i. Current readings shall be recorded from DCS/PLC (In case of non-availability of current readings in PLC/DCS, it should be recorded as an exception).
- ii. Belt Weigher Readings for flow rate shall be taken at intervals of every 5

minutes.

- iii. All data of capacity testing to be recorded as per Annexure-I

6.4 Performance parameters of various equipment/ system for demonstration

- i. Conveyor system: There should be no roll back of stopped conveyors fully or partly loaded. There shall be no buildup of coal in chutes, spillage of coal from skirt/conveyor/hoods while the belt is running.
- ii. Coasting Time: Coasting down time of conveyors shall be observed. There shall not be any buildup of coal in chutes after the stoppage of conveyor.
- iii. Belt Alignment: To check against any sway during operation under different loadings and different conveyor criss-cross combinations.
- iv. Self-aligning idlers: To check for satisfactory performance of self-aligning idlers.
- v. Control/Sequential Operation: Performance checks for specified controls including sequential starting/ stopping of conveyors and other equipment as per approved control write- up. Satisfactory operation of PLC and control systems including absence of spurious trips/communication failures of PLC to be demonstrated. Demonstration for detection of operated pull cord/belt sway switch along with display of conveyor number and operated pull cord/belt sway number.
- vi. Protection Switches: Satisfactory operation of pull chord, belt sway, zero speed, chute blockage and other safety / protection switches.
- vii. Load Start: Demonstration of the ability of conveyor to start under fully designed Load.
- viii. Gear Box: Temperature not to exceed design values. No abnormal sound and leakages should be observed. Satisfactory performance of hold back devices in preventing running back of conveyor belts in case of conveyor being stopped in fully loaded (at design load) condition to be checked.
- ix. Fluid Coupling: No excessive rise of temperature. Satisfactory operation of scoop in remote modes
- x. Brakes: Satisfactory operation to avoid buildup of coal in chutes.
- xi. Stacker Reclaimer:
 - a. Stacking at maximum boom height.
 - b. Starting of Boom Conveyor at full load and maximum inclination.
 - c. Reclaiming at maximum and minimum stockpile heights d.
 - Uniform to and fro travel speed and proper reeling / unreeling of flexible cables.
 - e. Satisfactory operation of all limit switches, protections, and other controls.
 - f. No sway of Boom Conveyor during stacking and reclaiming a t different loads and inclinations.
 - g. Proper hoisting / luffing operation without jerks and operator cabin level adjustment.
 - h. No leakage from hydraulic couplings, seals, heat exchangers, instrument tapings etc.

- i. Satisfactory performance of all thruster brakes and rail clamps
 - j. Auto lubrication system should be in service.
- xii. Travelling Trippers
 - a. Check for smooth to & fro travel.
 - b. Operation of brakes and rail clamps functioning at full load.
 - c. Check against dragging during start up or running.
 - d. Check against blockage of discharge chutes at full load.
 - e. Check for end limit switches – position & functioning.
 - f. Check for proper festoon trolley operation/drag chain system.
 - g. Check for functioning of overspeed protection.
- xiii. Paddle Feeders
 - a. The paddle feeders should be able to move in both to and fro directions while feeding coal unto rated capacity without slippage of wheels and spillage of coal.
 - b. Demonstration of maximum & variable capacity.
 - c. Proper reeling and un-reeling operation of flexible cable. d. Proper position & functioning of end limit switches and anti collision devices.
- xiv. Crushers
 - a. Demonstration of maximum capacity.
 - b. Output coal size of (-) 20 mm coal to be ensured.
 - c. Vibration (velocity and amplitude) to be recorded at bearings.
 - d. No excessive rise in the temperature of bearings.
 - e. Satisfactory operation of Scoop mechanism from remote.
- xv. Vibrating Feeders/Vibrating Screen Feeders
 - a. Demonstration of rated capacity and variable capacity (if applicable).
 - b. Checking of stroke length and angle (as applicable)
- xvi. Wagon Tipplers
 - a. Smooth operation of wagon tipplers without jerks.
 - b. Demonstration of Capacity of unloading of wagon in terms of load, tipping rate, and complete unloading.
- xvii. Apron Feeders: Demonstration of satisfactory operation with variable Capacity adjustment & observation of vibration.
- xviii. Side Arm Chargers:
 - a. Demonstration of haulage capacity.
 - b. Smooth operation of sidearm charger / car puller / beetle charger

without jerks.

- c. Demonstration of placement of Loaded wagon & removal of empty wagon from tippler table.
- xix. Metal Detector: Tests for detection of specified smallest size pieces of go/ no go pieces of different metals (sphere/ cube). Go piece size less than 25mm for textile belting and go piece size less than 35 mm for steel cord belting. Operation of sand bag/bag/another marker/ counter to be checked.
- xx. In Line Magnetic Separator: Satisfactory Performance of magnet capacity to separate out specified sizes and weights of magnetic pieces buried under (rated load) coal. Proper discharge of picked up metal pieces into tramp metal chute with no coal pieces going into chute.
- xxi. Flap Gates: Satisfactory operation of remote, local, and manual actuation of gates. Check for stroke and operation of limit/torque switches.
- xxii. Belt Scrappers: Satisfactory performance of belt scrappers to be checked.
- xxiii. Dust Control System: Dust Extraction/Dust Suppression systems to be checked for satisfactory performance.

7.0 FREQUENCY OF CAPACITY TESTING

Capacity Test shall be conducted three times a year. One in First Quarter (April-May), second in Third Quarter (November-December) and third in Last Quarter (January- March).

During Monsoon Period (Between Jun 1st – October 31st) capacity test may not be conducted due to the constraint of wet and sticky coal receipt.

8.0 RESPONSIBILITY OF THE TEST

EEMG of the station will be responsible for the compliance of the test every quarter. CHP (Operation, MM, EM & C&I) will be responsible for conducting the test.

RHQ-OS representative will associate in the capacity test being done by station once a year (any of the three quarter).

Station shall demonstrate to CC OS Representative, any one stream (any flow path described earlier in this document) capacity test during yearly technical audit. CC OS technical Audit Schedule shall be communicated minimum 07 days in advance for preparation.

9.0 REPORT

Based on the records of performance & capacity test as mentioned in point 6.3, if any deviation is found, an action plan to mitigate the same shall be prepared. Compliance of Performance and Capacity Test shall also be recorded in CHP ORT Sheet. (Refer Table-3 of Annexure-I).

Signed log sheets as per Annexure-I to be submitted to COS within 7 days of capacity testing.

Exceptions and actions taken to be monitored in monthly ORT through CHP ORT Sheet & also in CC OS & RHQ-OS monthly structured meetings.

10.0 RECORDS TO BE MAINTAINED

Sl No.	Record Description	Prepared by	Frequency	Remarks
1	Log of Capacity Test	CHP	Quarterly	SAMPLE RECORD SHARED
2	Exceptions/Deviations with action plan.	CHP	Quarterly	

ANNEXURE –I LOG SHEET (To be filled for each flow path)

Table- 1

Flow Path		(Example: 1A-2A-CR1&2-3A-Tripper)	
Belt Weigher Location			
Start Time		Belt Weigher Totalizer Reading at start	
End Time		Belt Weigher Totalizer Reading at end	
Duration of Test		Coal Conveyed (in MT)	
Rated Capacity		Average Feed Rate achieved (in MT)	

Table-2

Time (to be noted every 5 minutes)	Current Readings (in A)					Belt Weigher Reading (Conv....)
	Conv 1A	Conv 2A	CR1	CR2	Conv 3A	
Rated Current						NA
No Load Current						NA
Time+5 mins	*	*	*	*	*	
Time+10 mins	*	*	*	*	*	
...						

* Current measurement to be done once flow rate stabilizes.

Table-3

Capacity Deficiency/ Shortcomings/ Defects observed, if any:

Sl.No.	Name of Equipment	Deficiency/ Defects observed	Action Plan	Target Date

Sign of EEMG CHP EMD C&I

Sign of HOD CHP

Sign of HOD FM

GM O&M

SAMPLE CHP CAPACITY TEST RECORD

S. no.	Date / Time of Test	Stage	Flow Path	System In Service	Design TPH	Calculated TPH	Remarks
1.							
2.							
3.							
4.							
5.							

SAMPLE Remarks:

1. In Stage I Tipper chute was overflowing sometimes on full loads.
2. In Stg III conv 26A/B overload was occurring causing tripping of downstream conveyors. To be checked/rectified.
3. Conv#5A not moving in loaded condition.